USN | | | | | | | | | | | [22CE43]

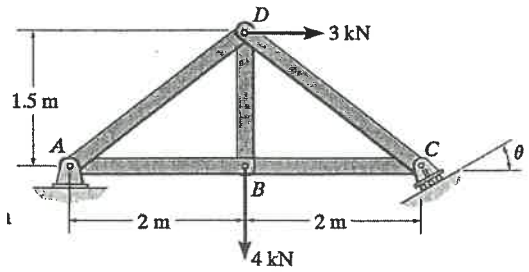
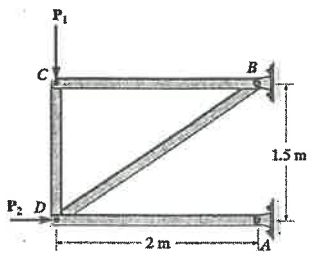
FOURTH SEMESTER B.Tech EXAMINATION, MAY - 2024

ANALYSIS OF DETERMINATE STRUCTURES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5 x2=10**
1.	**a.**	Define rigid or perfect truss with explaining the condition for collapsibility.	L1	CO1	**02**
	b.	State moment area theorems	L1	CO2	**02**
	c.	Define slenderness ratio and radius of gyration.	L1	CO3	**02**
	d.	Distinguish between two hinged and three hinged arches	L1	CO4	**02**
	e.	Explain the concept of influence lines	L1	CO5	**02**
		PART – B			**6 X 5=30**
2.	**a.**	Determine the force in each member of the truss, and state if the members are in tension or compression for truss Shown in Figure 2(a) θ= 0°.  **Figure 2(a)**	L3	CO1	**05**
		OR			
	b.	Determine the force in each member of the truss and state if the members are in tension or compression for truss Shown in Figure 2(b). Set P1 = 45 kN, P2 = 30 kN.  **Figure 2(b)**	L3	CO1	**05**
3.	**a.**	Derive differential deflection equation for elastic curve ($EI\ d^2y/dx^2=M$)	L3	CO2	**05**
		OR			
	b.	A beam carries 4 m long simply supported at its ends, carries a point load W at its centre. If the slope at the ends of the beam is not to exceed 1°, find the deflection at the centre of the beam	L3	CO2	**05**
4.	**a.**	The external and internal diameters of the hollow cast iron column are 5cm and 4cm respectively. If the length of this column is 3m and both	L3	CO3	**05**

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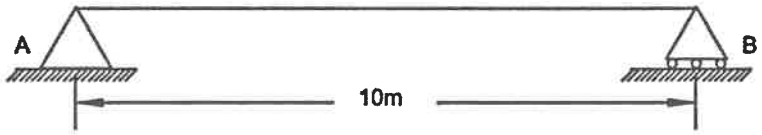
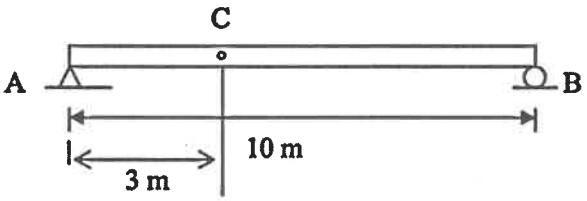
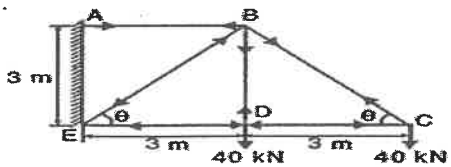
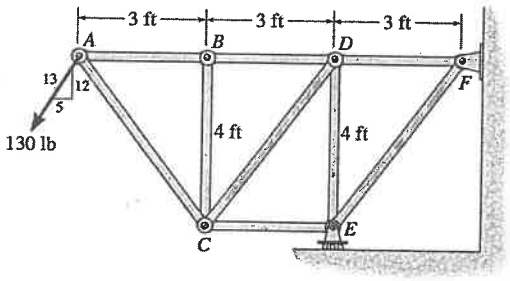
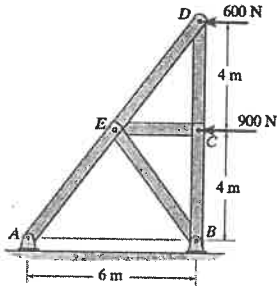
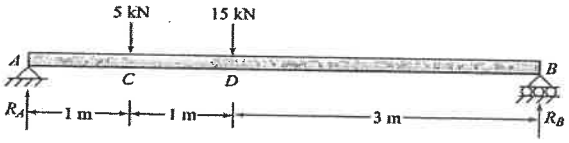
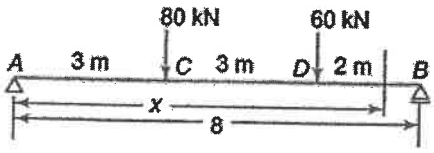
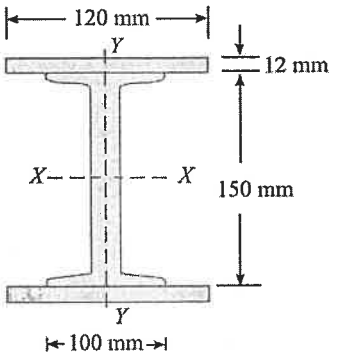
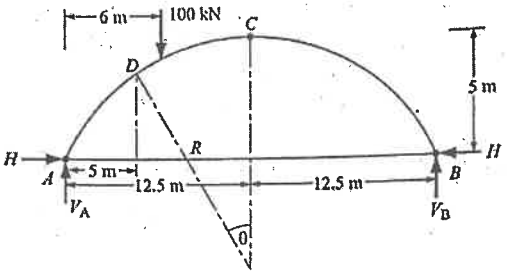
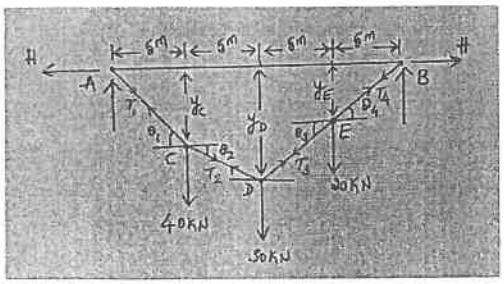
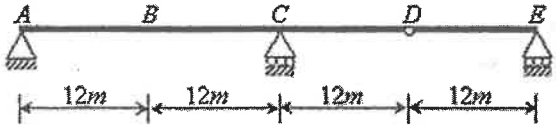
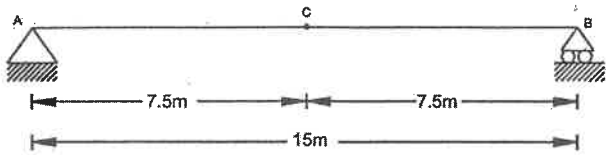
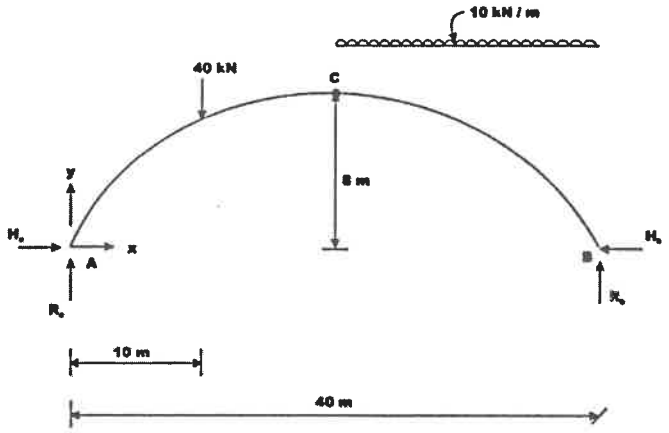
		of its ends are fixed, determine the crippling load using Rankine's formula. Take σ_c= 550N/mm^2 and a=1/1600			
		OR			
	b.	Compare the ratio of the strength of a solid steel column to that of a hollow of the same cross-sectional area. The internal diameter of the hollow column is 3/4 of the external diameter. Both the columns have the same length and are pinned at both ends.	L3	CO3	**05**
5.	**a.**	3-hinged arch has a span of 30m and a rise of 10m. The arch carries UDL of 0.6 kN/m on the left half of the span. It also carries 2 concentrated loads of 1.5 kN and 1.3 kN at 5 m and 10 m from the 'right' end. Determine the reactions at the support.	L3	CO4	**05**
		OR			
	b.	A circular arch to span 25 m with a central rise 5m is hinged at the crown and springing. It carries a point load of 100 kN at 6m from left support. Calculate 1. The reactions at supports 2. Moment at 5m from the left support	L3	CO4	**05**
6.	**a.**	Construct the influence line for the reaction at support B for the beam of span 10m. The beam structure is shown in Figure 6(a)  **Figure 6(a)**	L3	CO5	**05**
		OR			
	b.	Construct the influence lines for bending moment at C The beam structure is shown in Figure 6(b)  **Figure 6(b)**	L3	CO5	**05**
7.	**a.**	Find the forces in all the members of the truss shown in Fig 7(a). Tabulate the results  **Figure 7(a)**	L3	CO1	**05**
		OR			
	b.	A solid round bar of 60 mm diameter and 2.5 m long is used as a strut. Find the safe compressive load for the strut using Euler's formula if (a) both ends are hinged (b) both ends are fixed. Take E= 2 × 10^{5}N/mm^2 and factor of safety = 3.	L3	CO3	**05**
		PART – C			**6 X 10=60**

8.	**a.**	Determine the force in each member of the truss. State if the members are in tension or compression, truss shown in Figure 8(a)  **Figure 8(a)**	L3	CO1	**10**
		OR			
	b.	Determine the force in each member of the truss, and state if the members are in tension or compression, truss shown in Figure 8(b)  **Figure 8(b)**	L3	CO1	**10**
9.	**a.**	Two point loads of 5kN and 15kN are acting on a 5m simple beam shown in Figure 9(a) at 1m and 2 m respectively from the left end. Find the following: 1. The slopes at the 2 ends 2. The deflections under the applied loads 3. The position & magnitude of the maximum deflection. Take E= 90 GPa, I= $18*10^6$ mm^4  **Figure 9(a)**	L3	CO2	**10**
		OR			
	b.	A simply supported beam, 8 m long, carries two concentrated loads of 80 kN and 60 KN at a distance of 3m and 6m from the left end respectively. Calculate the slopes at the ends and the deflections under the loads. Take E= 200 GN/m2 and I= 30,000 cm4. Beam shown in Figure 9(b)  **Figure 9(b)**	L3	CO2	**10**
10.	**a.**	Compare the ratio of the strength of a solid steel column to that of a hollow of the same cross-sectional area. The internal diameter of the hollow column is 3/4 of the external diameter. Both the columns have the same length and are pinned at both ends.	L3	CO3	**10**

		OR			
	b.	Figure shows a built-up column consisting of 150 mm×100mm steel section. with 120 mm × 12 mm plate riveted to each flange. Calculate the safe load, the column can carry, if it is 4 m long having one end fixed and the other hinged with a factor of safety 3.5. Take the properties of the joist as Area = 2167 mm^2, I_{XX} = 8.391 × 106 mm^4, I_{YY} = 0.948 × 10^6 mm^4. Assume the yield stress as 315 MPa and Rankine's constant (a) = 1/7500 	L3	CO3	**10**
11.	**a.**	A circular arch to span 20 m with a central rise 5 m is hinged at the crown and springing. It carries a point load of 100 kN at 6 m from the left support. Calculate 1) The reactions at the supports, 2) The reactions at crown, 3) Moment at 5 m from the left support. Arch shown in Figure 11(a)  **Figure 11(a)**	L3	CO4	**10**
		OR			
	b.	A light cable is supported at two points 20m apart which are at the same level. The cable supports three concentrated loads as shown in the below figure 11(b). The deflection 'y_C' at first point is found to be 0.8m. Determine the tension in different segments and also total length of the cable  **Figure 11(b)**	L3	CO4	**10**
12.	**a.**	Construct the influence lines for reactions at A and E, also shear and	L3	CO5	**10**

		moment at D and B respectively beam shown in Figure 12(a)  **Figure 12(a)**			
		OR			
	b.	Construct the influence line for the moment at point C of the beam shown in Figure 12 (b)  **Figure 12 (b)**	L3	CO5	**10**
13.	**a.**	The hollow cylindrical cast iron column is 4m long with both ends fixed. Determine the minimum diameter of the column if it has to carry a safe load of 250KN with a FOS of 5 take internal diameter as 0.8 times the external diameter. Take σ_c= 550N/mm^2 and a=1/1600	L3	CO3	**10**
		OR			
	b.	A three-hinged parabolic arch is loaded as shown in Fig 13(b). Calculate the location and magnitude of maximum bending moment in the arch. Draw bending moment diagram  **Figure 13 (b)**	L3	CO4	**10**